Color naming.

Basic color free-elicitation:
English:

> Name 15 basic colors.

Russian:

> Перечислите 15 основных цветов.

Color naming elicitation:
English:

> Here is a list of 15 basic color names: <shuffled basic color list>.
> Which of these names best describes the following color: <hex-code>?
> Respond only using the name.

Russian:

> Вот список из 15 названий основных цветов: <shuffled basic color list> .
> Какое из названий цветов лучше всего описывает следующий цвет: <hex-code>?
> Отвечайте только названием одного цвета из списка.

We repeated this prompt ten times for each WCS color with the basic color list shuffled each time and temperature set to the default 0.7 to elicit ten names per WCS color. For each of the ten elicitations per color, if the output was not one of the 15 basic colors we would keep re-querying GPT until it did give a valid color output (GPT-4 is slightly non-deterministic even at temperature 0 due to changes in hardware). If after 10 attempts the response was still invalid, we would return "error" as the color (this response is later discarded from the analysis).

Additional details of behavioral experiments.

Similarity Experiments: Participants. To collect similarity judgments over pitch and vocal consonants we deployed two online experiments on Amazon Mechanical Turk (AMT).[**] The recruitment and experimental pipelines were automated using PsyNet (42), a modern framework for experiment design[††] and deployment which builds on the Dallinger[‡‡] platform for recruitment automation. Overall, 55 participants completed the pitch study and 64 participants completed the vocal consonants study. Participants were recruited from the United States, were paid $9-12 USD per hour, and provided informed consent as approved by the Princeton IRB (#10859) and the Max Planck Ethics Council (#2021_42).

To enhance data quality, participants had to pass a standardized headphone check (43) that ensures good listening conditions and task comprehension, and were required to have successfully completed at least 3000 tasks on AMT. Upon passing the prescreening stage, participants were randomly assigned to rate the similarity between different pairs of stimuli and provided numerical judgments on a 7-Likert scale ranging from 0 (completely dissimilar) to 6 (completely similar). In the pitch experiment, participants provided an average of 80 judgments, and in the vocal consonants experiment an average of 55 judgments. See SI for explicit instructions.

Similarity Experiments: Procedure. Upon providing informed consent and passing the headphone check, participants received the following instructions. In the case of the pitch experiment: "In this experiment we are studying how people perceive sounds. In each round you will be presented with two sounds and your task will be to simply judge how similar those sounds are. You will have seven response options, ranging from 0 ('Completely Dissimilar') to 6 ('Completely Similar'). Choose the one you think is most appropriate. You will also have access to a replay button that will allow you to replay the sounds if needed. Note: no prior expertise is required to complete this task, just choose what you intuitively think is the right answer." Participants were then informed of an additional small quality bonus "The quality of your responses will be automatically monitored, and you will receive a bonus at the end of the experiment in proportion to your quality score. The best way to achieve a high score is to concentrate and give each round your best attempt". While the task is subjective in nature, we used consistency as a proxy for quality by repeating 5 random trials at the end of the experiment and computing the Spearman correlation s between the original responses and their repetitions. The final bonus was computed using the formula $\min(\max(0.0, 0.1s), 0.1)$ yielding at most 10 cents. In the main experiment participants were assigned to random stimulus pairs and were instructed to rate their similarity using the following prompt: "How similar are the pair of sounds you just heard?" and provided a response on a Likert scale. The procedure for the vocal consonants similarity experiment was identical up to the specific instructions. Specifically, participants received the following instructions: "In this experiment we are studying how people perceive the sound of vocal consonants. A consonant is a speech sound that is pronounced by partly blocking air from the vocal tract. For example, the sound of the letter *c* in *cat* is a consonant, and so is *t* but not *a*. Similarly, the sound of the combination *sh* in *sheep* is a consonant, and so is *p* but not *ee*. In general, vowel sounds like those of the letters *a, e, i, o, u* are not consonants." The instructions then proceeded: "In each round you will be presented with two different recordings each including one consonant sound and your task will be to simply judge how similar are the sounds of the two spoken consonants. We are not interested in the vowel sounds nor in the voice height, just the sound of the consonants. You will have seven response options, ranging from 0 ('Completely Dissimilar') to 6 ('Completely Similar'). Choose the one you think is most appropriate. Note: no prior expertise is required to complete this task, just choose what you intuitively think is the right answer." Participants were then informed of the quality bonus which was identical to the pitch task, and then rated the similarity between pairs of random consonants based on the following prompt "How similar is the sound of the consonants pronounced by the two speakers?" and a Likert scale as before.

[**] https://www.mturk.com/
[††] https://psynet.dev/
[‡‡] https://dallinger.readthedocs.io/

Similarity Experiments: Model Evaluation. We quantified model performance in predicting human similarity judgments by computing the Pearson correlation coefficient between the flattened upper triangle of the LLM-based and human-based similarity matrices (to account for the fact that these matrices are symmetric). This approach is similar to representational similarity analysis (44). To compute 95% confidence intervals, we bootstrapped with replacement over model predictions with 1,000 repetitions and computed for each repetition the average similarity matrix. We then correlated the upper triangles of each of those matrices with human data to produce a list of correlation coefficients on which we computed confidence intervals.

Color Naming Experiments: Participants. To collect the color naming data in Russian and British English participants, we ran online experiments on Prolific[§§]. Overall, we recruited 103 UK participants and 51 Russian participants. All texts in the interface of the experiment (e.g., buttons, instructions, etc) were presented in the native language of the participant. The Russian texts were first automatically translated using DeepL[¶¶] and then manually checked and corrected by a native speaker of Russian (author I.S). Participants had to be raised monolingually and to speak the target language as their mother tongue. Each participant was paid 9 GBP per hour and provided informed consent according to an approved protocol (Max Planck Ethics Council #2021_42). The experiment was implemented using PsyNet (42). Each session starts with a free-elicitation task where participants are asked to provide basic colors:

Color Naming Experiments: Procedure.

Basic color free-elicitation:

English:

> Please name at least 8 basic color names.
> Press enter after each color name.
> Only use lower-case letters.

Russian:

> Укажите не менее 8 названий основных цветов.
> Нажмите клавишу Enter после каждого названия цвета.
> Используйте только строчные буквы.

Participants may only submit color names without spaces, numbers, or special characters and can only submit the page if they have provided at least eight names. The list of obtained colors is highly overlapping with the GPT-4 list, justifying our choice to use GPT-4 as the basis for the word naming task (colors are sorted by their naming frequency).

Top 15 terms in English:

> "blue", "green", "yellow", "red", "purple", "orange", "black", "pink", "white", "brown", "grey", "violet", "indigo", "turquoise", "silver"

Top 15 terms in Russian:

> "красный", "синий", "белый", "зеленый", "оранжевый", "желтый", "фиолетовый", "черный", "голубой", "коричневый", "розовый", "серый", "жёлтый", "зелёный", "чёрный"

From the top 15 color terms, 11 (English) and 12 (Russian) color terms are overlapping with the list provided by GPT-4. Before the main experiments, participants received the following instructions:

Color naming instructions:

English:

> During this experiment, you will be presented with a square of a particular color and will be required to select the most suitable color term from a list of options.
> Please be aware that some of the colors may be repeated to verify consistency of your choices.
> If we detect any inconsistencies in your answers, we may terminate the experiment prematurely.
> The best strategy is to answer each question truthfully, as attempting to memorize responses may prove difficult.

Russian:

> В ходе этого эксперимента вам будет представлен квадрат определенного цвета, и вам нужно будет выбрать наиболее подходящее название цвета из списка вариантов.
> Имейте в виду, что некоторые цвета могут повторяться, чтобы убедиться в согласованности вашего выбора.
> Если мы обнаружим какие-либо несоответствия в ваших ответах, мы можем досрочно прекратить эксперимент.
> Лучшая стратегия - отвечать на каждый вопрос правдиво, так как попытка запомнить ответы может оказаться сложной.

The participants then went through the main experiment:

Color naming task:

English:

> <square of a particular color>
> You will see below a list of 15 basic color names. Which of these names best describes the color above?
> <shuffled basic color list presented as buttons>

Russian:

> <square of a particular color>
> Ниже Вы увидите список из 15 основных названий цветов.
> Какое из этих названий лучше всего описывает вышеуказанный цвет?
> <shuffled basic color list presented as buttons>

At the end of the experiment, the participant took a color blindness test (45). Some participants abandoned the experiment prematurely, but we nevertheless included their responses (42 English participants, 3 Russian participants). Only a fraction of the participants failed the color blindness test (5 of 103 English participants, and 2 of 51 Russian participants). Consistent with the WCS we included all participants including those who failed the color blindness test. In a control analysis, we excluded all color blind individuals and all participants that did not complete the entire session and got nearly identical results (the adjusted Rand index was 0.92 for English experiments and 0.97 for Russian experiments).

Color Naming Experiments: Analysis. For each color, we collected at least 10 responses per LLM variant, and at least 10 forced-choice human selections per color (English mean 19.30 responses, Russian mean 12.17 responses). Consistent with previous literature for each Munsell's color we selected the most frequently reported term. We then presented the dominant colors in Figure 3B. To aid visualization we average the RGB values of all colors with the same color term, and presented them as the legend and clustered color in that figure. We also listed per color the number of Munsell's colors that were associated with each dominant color term. Figure 3B provides additional information on the degree of agreement for each color. Colors for which less than 50% and 90% of the times the dominant color term was selected were indicated by "-" and "*", respectively. If the dominant color term was selected more than 90% of the time, no marking was used.

[§§] https://www.prolific.com
[¶¶] https://www.deepl.com

Adjusted Rand index. The Rand index (46) is a label-insensitive measure of clustering similarity that instead of relying on specific labels (e.g. "Blue") quantifies the similarity between two clustering partitions by counting pairs of items (in our case Munsell colors) that are clustered consistently and dividing them by the overall number of pairs. This allows to compare different clustering schemes when the vocabulary of labels is not aligned (e.g. English and Russian). Formally, we computed: $R = (b + c)/a$; where b is the number of pairs of items that are in the same subset in one clustering and in the same subset in the other, c is the number of pairs of items that are in different subsets in one clustering and in different subsets in the other and a is the total number of pairs. The Rand index provides high values for two random clusterings, to adjust for this we used the corrected-for-chance version of the Rand index (46), which normalizes the raw value by the expected value of the Rand index for random clusterings. Formally, we have $ARI = (RI - RI_{\text{rand}})/(1 - RI_{\text{rand}})$ where ARI is the adjusted Rand index, RI is the raw Rand index and RI_{rand} is the expected Rand index for random clusterings. The adjusted Rand index is thus ensured to have a value close to 0.0 for random labeling independently of the number of clusters and samples, exactly 1.0 when the clusterings are identical (up to a permutation), and reaches -0.5 for "orthogonal clusters" that are less consistent relative to what is expected by chance. In our case, all values were strictly positive suggesting consistency across languages and experiments. To compute confidence intervals, we created bootstrapped datasets by sampling the responses of each color with replacement and recomputing the dominant selected color name. We then obtained CIs by computing the adjusted Rand index for 1,000 pairs of bootstrapped datasets.

Lindsey and Brown dataset. We compared our experimental data to a dataset by (31), reproduced with permission by Delwin Lindsey. The data contains two experimental conditions conducted in the lab with the same 51 participants. In the first condition, participants were instructed to provide free naming responses. In the second condition, participants were instructed to choose from a pre-specified list of 11 color terms: (Green, Blue, Purple, Pink, White, Brown, Orange, Yellow, Red, Black, and Gray). Despite the fact that the Lindsey & Brown experiment was conducted in the lab (and not online like our experiments) and that the constrained list in our experiment was somewhat different, the results of both experiments were highly consistent with our human English data (Constrained, $ARI = 0.73$ 95% CI $[0.65, 0.74]$, free naming 0.75 95% CI $[0.66, 0.75]$). In addition to putting an upper bound on the consistency with which the LLM can predict human data (by comparing it with another human experiment), these results prove that despite less control over color presentation compared to the lab, online presentation still provides high-quality color naming data.